

Neo4j Hands-on Learning Component

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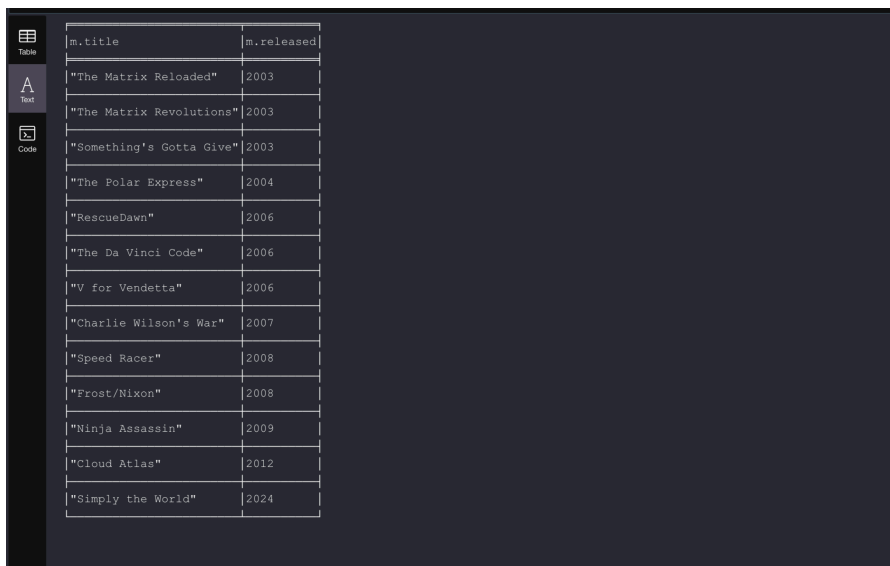
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Section : E

Part 1:- Movie Dataset

1. List all movies released after the year 2000.

A screenshot of a Neo4j browser interface. On the left, there is a sidebar with three icons: a table icon labeled 'Table', a text icon labeled 'Text', and a code icon labeled 'Code'. The 'Table' icon is selected. The main area displays a table with two columns: 'm.title' and 'm.released'. The table contains 15 rows of movie data.

m.title	m.released
"The Matrix Reloaded"	2003
"The Matrix Revolutions"	2003
"Something's Gotta Give"	2003
"The Polar Express"	2004
"Rescue Dawn"	2006
"The Da Vinci Code"	2006
"V for Vendetta"	2006
"Charlie Wilson's War"	2007
"Speed Racer"	2008
"Frost/Nixon"	2008
"Ninja Assassin"	2009
"Cloud Atlas"	2012
"Simply the World"	2024

2. Find all actors who acted in "The Matrix".

a.name
"Emil Eifrem"
"Hugo Weaving"
"Laurence Fishburne"
"Carrie-Anne Moss"
"Keanu Reeves"

3. Return all movies where “Tom Hanks” acted along with the roles he played.

m.title	r.roles
"Apollo 13"	["Jim Lovell"]
"You've Got Mail"	["Joe Fox"]
"A League of Their Own"	["Jimmy Dugan"]
"Joe Versus the Volcano"	["Joe Banks"]
"That Thing You Do"	["Mr. White"]
"The Da Vinci Code"	["Dr. Robert Langdon"]
"Cloud Atlas"	["Zachry", "Dr. Henry Goose", "Isaac Sachs", "Dermot Hoggins"]
"Cast Away"	["Chuck Noland"]
"The Green Mile"	["Paul Edgecomb"]
"Sleepless in Seattle"	["Sam Baldwin"]
"The Polar Express"	["Hero Boy", "Father", "Conductor", "Hobo", "Scrooge", "Santa Claus"]
"Charlie Wilson's War"	["Rep. Charlie Wilson"]

4. Find all directors who directed more than one movie.

d.name	moviesDirected
"Lana Wachowski"	5
"Lilly Wachowski"	5
"Rob Reiner"	3
"Ron Howard"	3
"Nora Ephron"	2
"Mike Nichols"	2
"James Marshall"	2
"Robert Zemeckis"	2

- Return the number of movies released each year, sorted in descending order by count.

year	movieCount
1999	4
1992	4
2003	3
2000	3
1998	3
1996	3
2006	3
1997	2
1986	2
1995	2
2008	2
1993	1
1990	1
2012	1

- Find all pairs of actors who have co-acted in at least one movie.
(Hint: use `MATCH (a1)-[:ACTED_IN]->(m)<-[:ACTED_IN]-(a2)` and avoid duplicates.)

Actor1	Actor2	Movies
"Hugo Weaving"	"Emil Eifrem"	["The Matrix"]
"Laurence Fishburne"	"Emil Eifrem"	["The Matrix"]
"Carrie-Anne Moss"	"Emil Eifrem"	["The Matrix"]
"Keanu Reeves"	"Emil Eifrem"	["The Matrix"]
"Laurence Fishburne"	"Hugo Weaving"	["The Matrix", "The Matrix Reloaded", "The Matrix Revolutions"]
"Carrie-Anne Moss"	"Hugo Weaving"	["The Matrix", "The Matrix Reloaded", "The Matrix Revolutions"]
"Keanu Reeves"	"Hugo Weaving"	["The Matrix", "The Matrix Reloaded", "The Matrix Revolutions"]
"Carrie-Anne Moss"	"Laurence Fishburne"	["The Matrix", "The Matrix Reloaded", "The Matrix Revolutions"]
"Keanu Reeves"	"Laurence Fishburne"	["The Matrix", "The Matrix Reloaded", "The Matrix Revolutions"]
"Keanu Reeves"	"Carrie-Anne Moss"	["The Matrix", "The Matrix Reloaded", "The Matrix Revolutions"]
"Charlize Theron"	"Al Pacino"	["The Devil's Advocate"]
"Keanu Reeves"	"Al Pacino"	["The Devil's Advocate"]
"Keanu Reeves"	"Charlize Theron"	["The Devil's Advocate"]
"Kevin Pollak"	"James Marshall"	["A Few Good Men"]
"J.T. Walsh"	"James Marshall"	["A Few Good Men"]

- List all movies that have both "Keanu Reeves" and "Laurence Fishburne" in the cast.

Table	m.title
Text	"The Matrix Revolutions"
Code	"The Matrix Reloaded"
	"The Matrix"

- Find the average number of roles per actor.

Table	averageRolesPerActor
Text	1.6862745098039222
Code	

9. Return all people who both acted and directed in at least one movie.

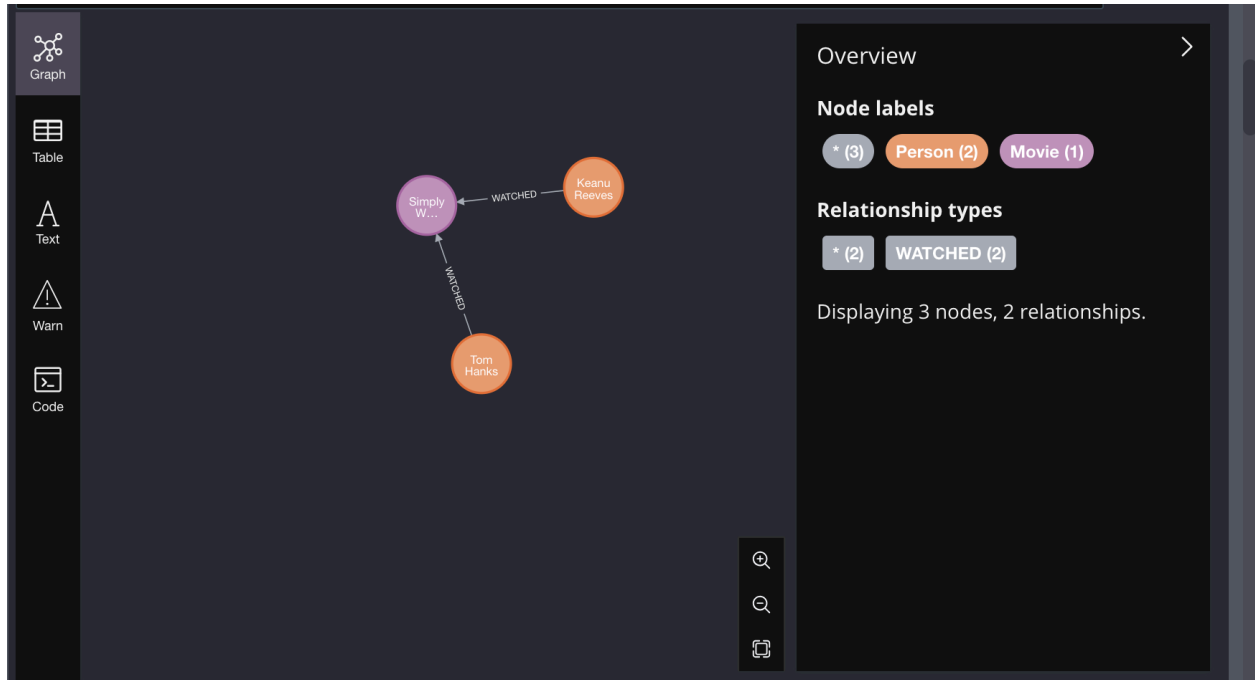
Table	p.name
Text	"Tom Hanks"
Code	"Werner Herzog"
	"Clint Eastwood"
	"James Marshall"
	"Danny DeVito"

10. List the top 5 actors who acted in the most movies.

a.name	moviesCount
"Tom Hanks"	12
"Keanu Reeves"	7
"Jack Nicholson"	5
"Hugo Weaving"	5
"Meg Ryan"	5

11. Find the shortest connection path (any relationship) between “Tom Hanks” and “Keanu Reeves”.

path
(:Person {born: 1956,name: "Tom Hanks"})-[:WATCHED]->(:Movie {tagline: "Beauty in Simplicity",title: "Simply the World",released: 2024})<-[:WATCHED]-(:Person {born: 1964,name: "Keanu Reeves"})



12. For each movie, return the number of unique people involved (actors, directors, producers, writers).

m.title	peopleInvolved
"A Few Good Men"	13
"Jerry Maguire"	10
"Speed Racer"	10
"Cloud Atlas"	9
"V for Vendetta"	9
"The Green Mile"	9
"Ninja Assassin"	8
"The Matrix"	8
"Top Gun"	8
"Stand By Me"	8
"The Matrix Reloaded"	7
"The Matrix Revolutions"	7
"You've Got Mail"	7
"Sleepless in Seattle"	7
"A League of Their Own"	7
"Frost/Nixon"	6

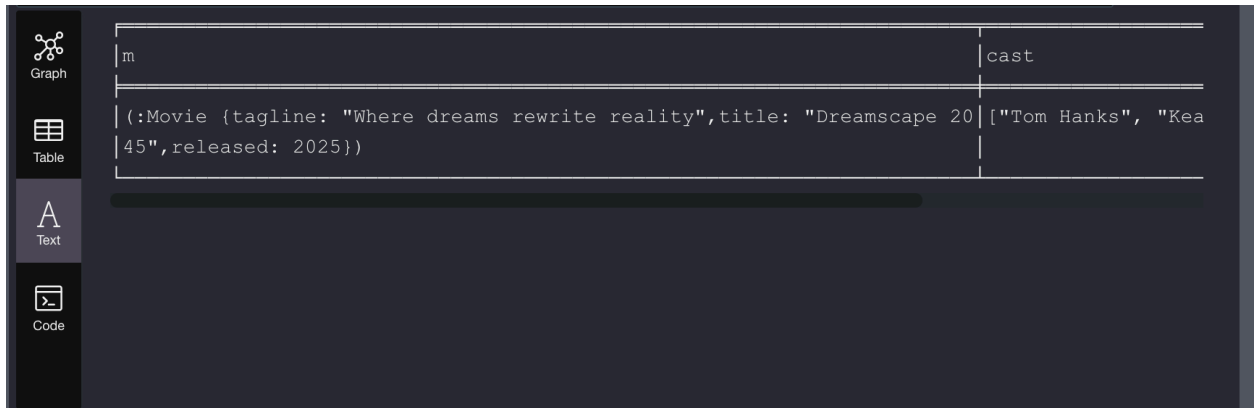
"The Matrix Reloaded"	7
"The Matrix Revolutions"	7
"You've Got Mail"	7
"Sleepless in Seattle"	7
"A League of Their Own"	7
"Frost/Nixon"	6
"Apollo 13"	6
"What Dreams May Come"	6
"When Harry Met Sally"	6
"As Good as It Gets"	5
"Snow Falling on Cedars"	5
"The Replacements"	5
"RescueDawn"	5
"Johnny Mnemonic"	5
"The Da Vinci Code"	5
"Twister"	5

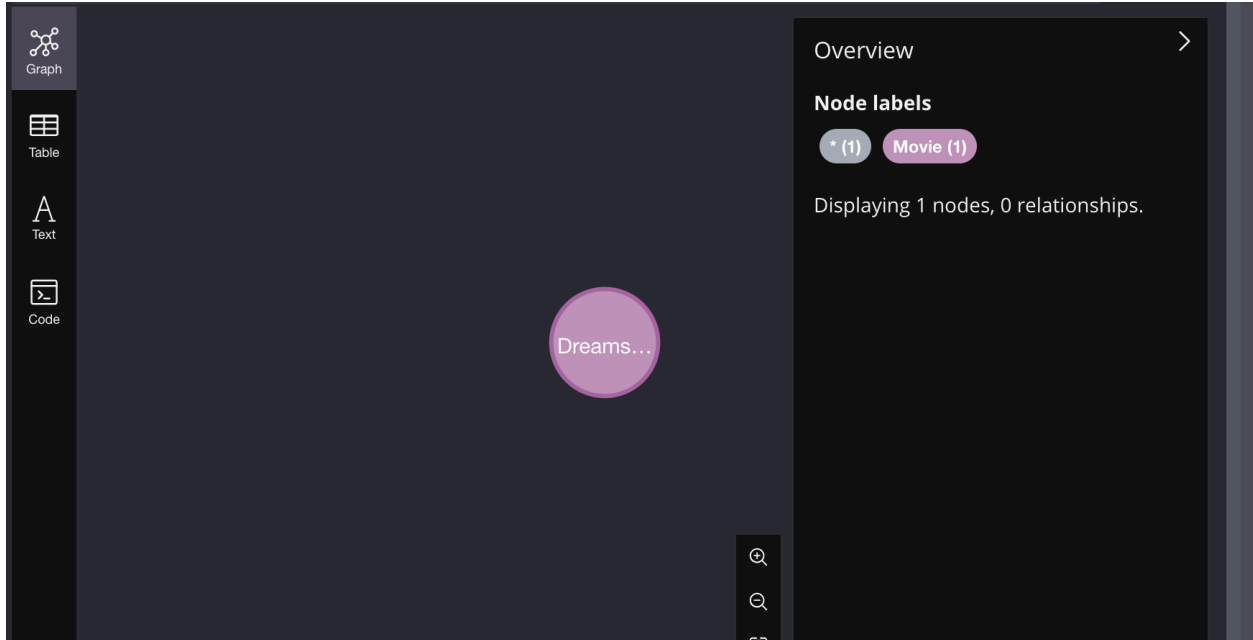
"RescueDawn"	5
"Johnny Mnemonic"	5
"The Da Vinci Code"	5
"Twister"	5
"The Birdcage"	4
"Hoffa"	4
"Something's Gotta Give"	4
"Charlie Wilson's War"	4
"The Devil's Advocate"	4
"Joe Versus the Volcano"	4
"That Thing You Do"	3
"Unforgiven"	3
"Cast Away"	3
"One Flew Over the Cuckoo's Nest"	3
"Bicentennial Man"	3
"The Polar Express"	2

13. Find actors who have only acted in movies directed by “Christopher Nolan”.



14. Create a new node for a fictional movie and link at least 3 existing actors as **ACTED_IN**.





15. Find triads of actors who have co-starred together in at least two different movies.

["Tom Hanks", "Meg Ryan", "Steve Zahn"]	["Joe Versus the Volcano", "Sleepless in Seattle"]
["Tom Hanks", "Meg Ryan", "Dave Chappelle"]	["Joe Versus the Volcano", "Sleepless in Seattle"]
["Meg Ryan", "Tom Hanks", "Parker Posey"]	["Sleepless in Seattle", "Joe Versus the Volcano"]
["Meg Ryan", "Tom Hanks", "Greg Kinnear"]	["Sleepless in Seattle", "Joe Versus the Volcano"]
["Meg Ryan", "Tom Hanks", "Steve Zahn"]	["Sleepless in Seattle", "Joe Versus the Volcano"]
["Meg Ryan", "Tom Hanks", "Dave Chappelle"]	["Sleepless in Seattle", "Joe Versus the Volcano"]
["Meg Ryan", "Tom Hanks", "Victor Garber"]	["You've Got Mail", "Joe Versus the Volcano"]
["Meg Ryan", "Tom Hanks", "Bill Pullman"]	["You've Got Mail", "Joe Versus the Volcano"]
["Meg Ryan", "Tom Hanks", "Rita Wilson"]	["You've Got Mail", "Joe Versus the Volcano"]
["Meg Ryan", "Tom Hanks", "Rosie O'Donnell"]	["You've Got Mail", "Joe Versus the Volcano"]
["Tom Hanks", "Meg Ryan", "Victor Garber"]	["You've Got Mail", "Joe Versus the Volcano"]
["Tom Hanks", "Meg Ryan", "Bill Pullman"]	["You've Got Mail", "Joe Versus the Volcano"]
["Tom Hanks", "Meg Ryan", "Rita Wilson"]	["You've Got Mail", "Joe Versus the Volcano"]
["Tom Hanks", "Meg Ryan", "Rosie O'Donnell"]	["You've Got Mail", "Joe Versus the Volcano"]
["Tom Hanks", "Meg Ryan", "Nathan Lane"]	["You've Got Mail", "Sleepless in Seattle"]
["Meg Ryan", "Tom Hanks", "Nathan Lane"]	["Sleepless in Seattle", "You've Got Mail"]

Table	triad	movies
Text	["Hugo Weaving", "Laurence Fishburne", "Emil Eifrem"]	["The Matrix Revolutions", "The Matrix Reloaded"]
Code	["Hugo Weaving", "Laurence Fishburne", "Carrie-Anne Moss"]	["The Matrix Revolutions", "The Matrix Reloaded", "The Matrix"]
	["Hugo Weaving", "Laurence Fishburne", "Keanu Reeves"]	["The Matrix Revolutions", "The Matrix Reloaded", "The Matrix"]
	["Hugo Weaving", "Carrie-Anne Moss", "Emil Eifrem"]	["The Matrix Revolutions", "The Matrix Reloaded"]
	["Hugo Weaving", "Carrie-Anne Moss", "Laurence Fishburne"]	["The Matrix Revolutions", "The Matrix Reloaded", "The Matrix"]
	["Hugo Weaving", "Carrie-Anne Moss", "Keanu Reeves"]	["The Matrix Revolutions", "The Matrix Reloaded", "The Matrix"]
	["Hugo Weaving", "Keanu Reeves", "Emil Eifrem"]	["The Matrix Revolutions", "The Matrix Reloaded"]
	["Hugo Weaving", "Keanu Reeves", "Laurence Fishburne"]	["The Matrix Revolutions", "The Matrix Reloaded", "The Matrix"]
	["Hugo Weaving", "Keanu Reeves", "Carrie-Anne Moss"]	["The Matrix Revolutions", "The Matrix Reloaded", "The Matrix"]
	["Laurence Fishburne", "Hugo Weaving", "Emil Eifrem"]	["The Matrix Revolutions", "The Matrix Reloaded"]
	["Laurence Fishburne", "Hugo Weaving", "Carrie-Anne Moss"]	["The Matrix Revolutions", "The Matrix Reloaded", "The Matrix"]
	["Laurence Fishburne", "Hugo Weaving", "Keanu Reeves"]	["The Matrix Revolutions", "The Matrix Reloaded", "The Matrix"]
	["Laurence Fishburne", "Carrie-Anne Moss", "Emil Eifrem"]	["The Matrix Revolutions", "The Matrix Reloaded"]
	["Laurence Fishburne", "Carrie-Anne Moss", "Hugo Weaving"]	["The Matrix Revolutions", "The Matrix Reloaded", "The Matrix"]
	["Laurence Fishburne", "Carrie-Anne Moss", "Keanu Reeves"]	["The Matrix Revolutions", "The Matrix Reloaded", "The Matrix"]
	["Laurence Fishburne", "Keanu Reeves", "Emil Eifrem"]	["The Matrix Revolutions", "The Matrix Reloaded"]

MAX COLUMN WIDTH:

Part 2:- Airport Routes Dataset

1. Project the Airport Routes graph into GDS in-memory graph.

nodeProjection	relationshipProjection	graphName
{Airport: {label: "Airport", properties: {}}}	{HAS_ROUTE: {aggregation: "DEFAULT", orientation: "UNDIRECTED", indexI nverse: false, properties: {distance: {aggregation: "DEFAULT", propert y: "distance", defaultValue: null}}, type: "HAS_ROUTE"}}	"airportGraph"

nodeCount	relationshipCount	projectMillis
3503	92778	8841

- Find the top 10 airports with the most connections (degree centrality).

airportCode	airportName	degree
"FRA"	"Frankfurt am Main"	610.0
"CDG"	"Paris Charles de Gaulle"	584.0
"IST"	"Istanbul International Airport"	575.0
"AMS"	"Amsterdam Airport Schiphol"	562.0
"MUC"	"Munich International Airport"	535.0
"ORD"	"Chicago O'Hare International Airport"	521.0
"DFW"	"Dallas/Fort Worth International Airport"	499.0
"DXB"	"Dubai International Airport"	484.0
"ATL"	"Hartsfield - Jackson Atlanta International Airport"	484.0
"PEK"	"Beijing Capital International Airport"	469.0

- Compute the average degree of all airports.

Table	averageDegree
Text	26.48529831572936
Code	

- Find the 5 airports with the lowest number of connections.

Table	airportCode	airportName	degree
Text	"TWB"	"Toowoomba Airport"	0.0
Code	"TXL"	"Berlin, Tegel International Airport *Closed*"	0.0
	"SXF"	"Berlin-Schönefeld International Airport *Closed*"	0.0
	"ILG"	"New Castle Airport"	0.0
	"CCV"	"Craig Cove Airport"	0.0

- Find the top 10 most influential airports using PageRank.

iata	airport_name	score
"IST"	"Istanbul International Airport"	16.086985682362783
"DFW"	"Dallas/Fort Worth International Airport"	15.812729757359648
"ORD"	"Chicago O'Hare International Airport"	14.855918386909076
"DEN"	"Denver International Airport"	14.164540660173802
"DXB"	"Dubai International Airport"	13.813150772061032
"PEK"	"Beijing Capital International Airport"	13.752834136304756
"ATL"	"Hartsfield - Jackson Atlanta International Airport"	13.579259790194984
"CDG"	"Paris Charles de Gaulle"	13.483728139313586
"FRA"	"Frankfurt am Main"	13.446623955873376
"LAX"	"Los Angeles International Airport"	12.373054606658952

- Persist PageRank scores back to the database as a property on Airport nodes.

```

1 CALL gds.pageRank.write('airportGraph', {
2   writeProperty: 'pagerank'
3 })
4 YIELD nodePropertiesWritten, ranIterations;
5
6 MATCH (a:Airport)
7 RETURN a.iata AS IATA, a.name AS Name, a.pagerank AS PageRank
8 ORDER BY a.pagerank DESC
9 LIMIT 10;

```

Table	IATA	Name	PageRank
Text	"IST"	null	16.08698568236279
Code	"DFW"	null	15.812729757359651
	"ORD"	null	14.855918386909083
	"DEN"	null	14.164540660173822
	"DXB"	null	13.813150772061034
	"PEK"	null	13.752834136304749
	"ATL"	null	13.579259790194984
	"CDG"	null	13.483728139313584
	"FRA"	null	13.446623955873376
	"LAX"	null	12.373054606658954

7. Identify all airports in the same connected component as JFK.

IATA
"AAA"
"AAE"
"AAL"
"AAN"
"AAQ"
"AAR"
"AAX"
"AAY"
"ABA"
"ABB"
"ABD"
"ABE"
"ABI"

"ABJ"
"ABL"
"ABM"
"ABQ"
"ABR"
"ABS"
"ABT"

JFK is part of a large connected component that includes approximately 1,000 other airports reachable through the HAS_ROUTE relationships.

- Count the total number of connected components in the airport network.

totalComponents
212

Started streaming 1 records after 1 ms and completed after 23 ms.

- Find the shortest route (by distance) between ATL and LHR.

totalCost	route
4197.0	["ATL", "DCA", "EWR", "LHR"]

10. Find the distance of the shortest path between JFK and CDG.

totalCost
3622.0

11. Identify the top 10 airports acting as transit hubs using Betweenness Centrality.

airport	score
null	390958.58509613085
null	368734.10410234286
null	365259.6507074962
null	340393.582883937
null	339441.8860074303
null	326830.23701284744
null	298891.74331870995
null	285270.75816178747
null	277537.0924865432
null	265922.7151575632

12. Find the 5 airports with the lowest Betweenness Centrality scores.

airport	score
"MCY"	0.0
"SAF"	0.0
"HRO"	0.0
"SLN"	0.0
"TXL"	0.0

13. Detect airport communities using Label Propagation.

communityCount	ranIterations
1286	4

14. Count the number of communities detected in the airport network.

numCommunities
1286

15. Find the top 5 airports most similar to CDG based on shared connections (Node Similarity).

Table	airport	similarity
Text	"FRA"	0.5
	"AMS"	0.4973958333333333
	"FCO"	0.4879518072289157
	"BRU"	0.44510385756676557
	"MUC"	0.44358974358974357
Code		